GWCD Documentation - Data Source

Vermont AHS GIS and Data Viz | Groundwater Chemistry Database (GWCD)

Data Source

Origin

Raw lab test results are submitted by Vermont-certified labs and land in a SQL Server table:

> `groundwater.ChemistryResults`

(accessed via `whEPHT_AHSSQLW01P_OS.sde` — an internal ArcGIS SDE connection)

This is the **system of record** for every analyte test submitted to the state, across all water source types (private well, spring, public/municipal, surface water, etc.) and all client types (homeowners, schools, daycares, VDH internal lab proficiency tests, etc.) — i.e. it is not pre-filtered for the private drinking water use case at all.

Weekly load

Every week, `GroundwaterChemistryDatabasePROD.py`:

1. Truncates and reloads a GIS-side staging table, `GroundwaterChemistry_GWCDrawdata`, with a 1:1 column copy of `groundwater.ChemistryResults`. This table is the full, **unfiltered, ungeocoded** snapshot of all submitted results — useful as a state-level dataset since it needs no geometry.
2. Runs weekly exceedance alerting (see below), cleaning/standardization, geocoding, exceedance calculation, and aggregation (see 02-etl-pipeline.md) against an in-memory copy of that table.
3. Writes two downstream products:
   - `GroundwaterChemistry_GWCDPoints` — filtered to the private drinking water scope, geocoded, with exceedance flags and per-town/analyte summary stats.
   - `GroundwaterChemistry_GWCDPointTableExternal` — the public-facing version of the above, deduplicated per location and suppressed for small sample sizes. This is what feeds the Experience Builder app.

Alias/reference tables used during cleaning

- `GroundwaterChemistryAliasTable` — maps problematic/misspelled town names (typos, out-of-state towns, placeholder values) to a canonical Vermont municipality, where a mapping exists.

- `GroundwaterChemistry_TownBounds` — Vermont town boundary polygons, used to spatially join each geocoded point to a `TOWNNAME`.

Weekly exceedance alert (separate from the public product)

Independent of the map/dashboard pipeline, the script also emails a rolling 7-day exceedance report (CSV + HTML table) to a fixed internal distribution list, classifying records into Metals / VOCs / BNAs / PCBs / Herbicides / PFOS-PFAS / Other. This uses its own, separate set of hardcoded concentration thresholds (see the script) and is **not** the same threshold table used for the public town-level exceedance rates — see 04-exceedance-calculations.md for that.